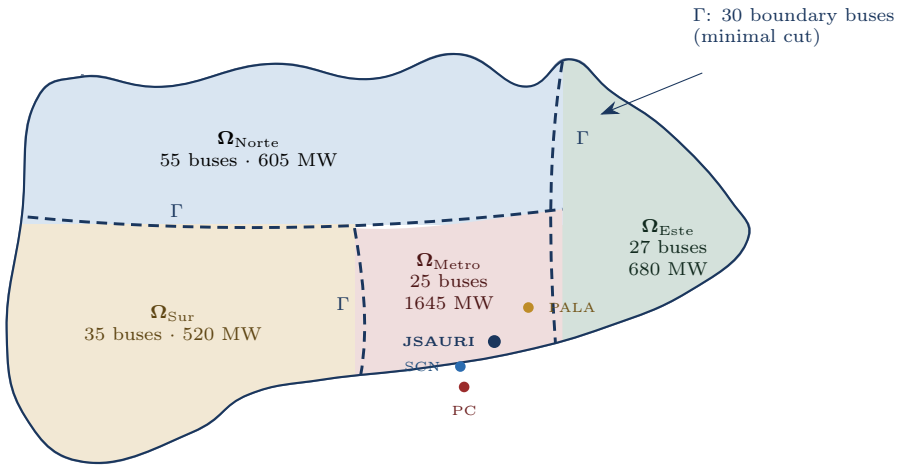
HBCF Diakoptic Condensation of the SENI-142 Benchmark

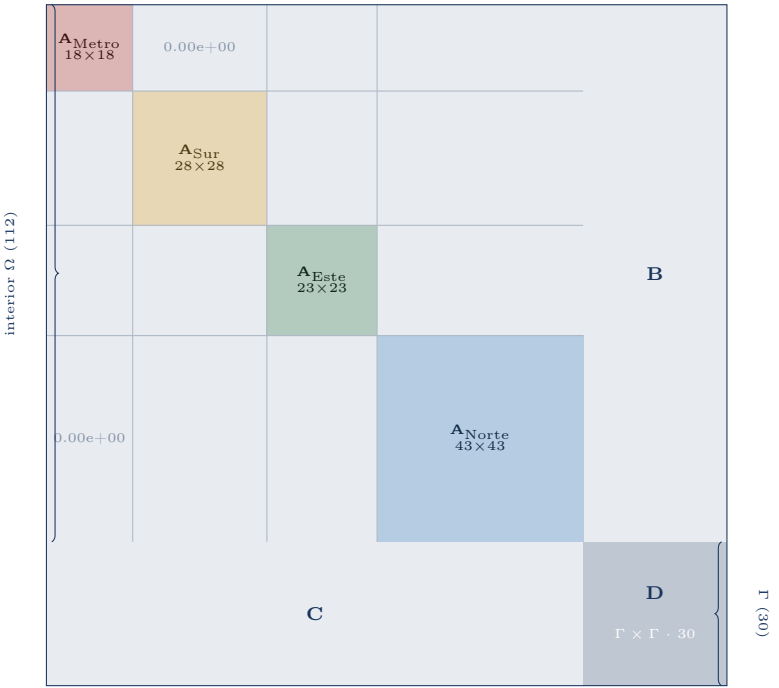
Four macro-domains with minimal boundary Γ and admittance block structure

(a) Island and boundary cut Γ



Σ = 142 buses · 3450.0 MW · block partition with minimal boundary

(b) Y = [A B; C D] — block structure



off-block ≡ 0.00e+00 (exact block-diagonality) · $\mathbf{S} = \mathbf{D} - \mathbf{C}\,\mathbf{A}^{-1}\mathbf{B}$

$\kappa_2(\mathbf{S}) = 2.49 \times 10^{3}$ · $\delta = 2.79 \times 10^{-14}$ pu (measured)